A quantum circuit for Shor's factoring algorithm using $2n+2$ qubits

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Abstract

- **We construct a quantum circuit for Shor's factoring algorithm using $2n+2$ qubits**
  - **[Qubit-efficient] The number of qubits is less than that in any other circuit ever constructed for the algorithm**
  - **[Size-efficient] The size of the circuit is about half that of Beauregard's circuit, which uses $2n+3$ qubits**

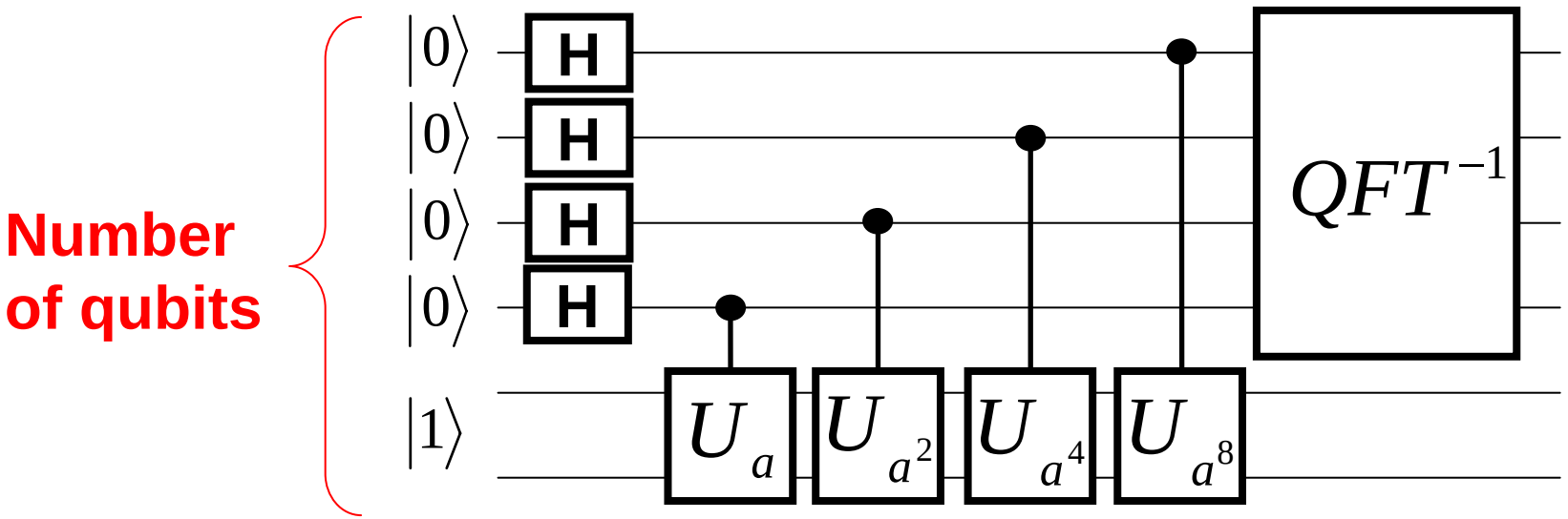
Background

- **Shor proposed an efficient quantum algorithm for factoring for which no efficient classical algorithm is known**
- **Efficient circuits for Shor's algorithm are useful for performing the algorithm on a quantum computer**

There is great interest in constructing efficient circuits for the algorithm

Circuits for Shor's algorithm

- **It suffices to construct a circuit for order-finding**



N : an n-bit number to be factored

a : a randomly chosen number less than N

U_a : a modular multiplication that maps $|x\rangle$ to $|ax \bmod N\rangle$

Previous circuits for order-finding

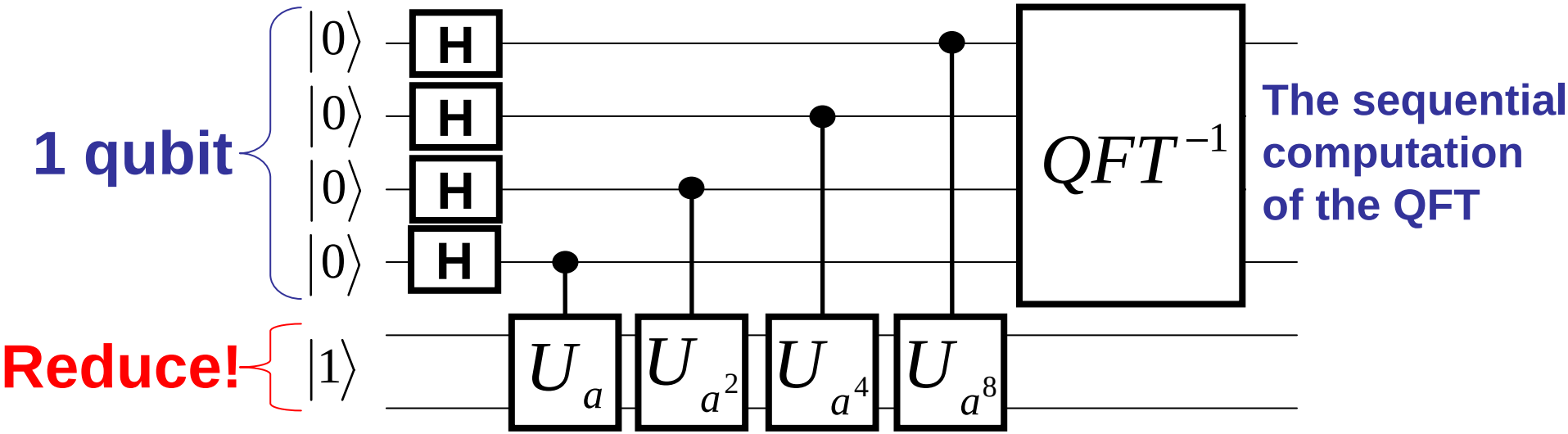
	Size	Depth	Number of qubits
Vedral et al. 1996	$O(n^3)$	$O(n^3)$	$3n+2$
Beauregard 2003	$O(n^3 \log n)$	$O(n^3)$	$2n+3$
Our circuit	$O(n^3 \log n)$	$O(n^3)$	$2n+2$

Half

1 qubit

The problem

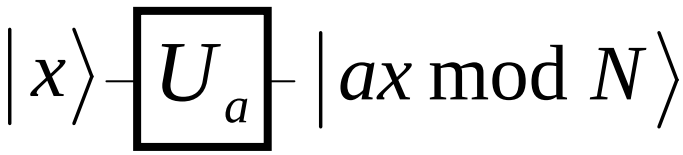
- **The problem is to reduce the number of qubits used for modular multiplication**



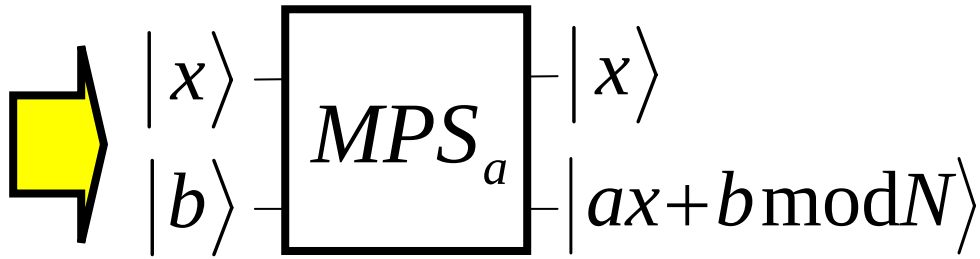
$2n$+2 qubits = 1 qubit + $2n$+1 qubits

Decomposition of modular multiplication

Modular multiplication



Modular product-sum



There are useful qubits for constructing the new circuit

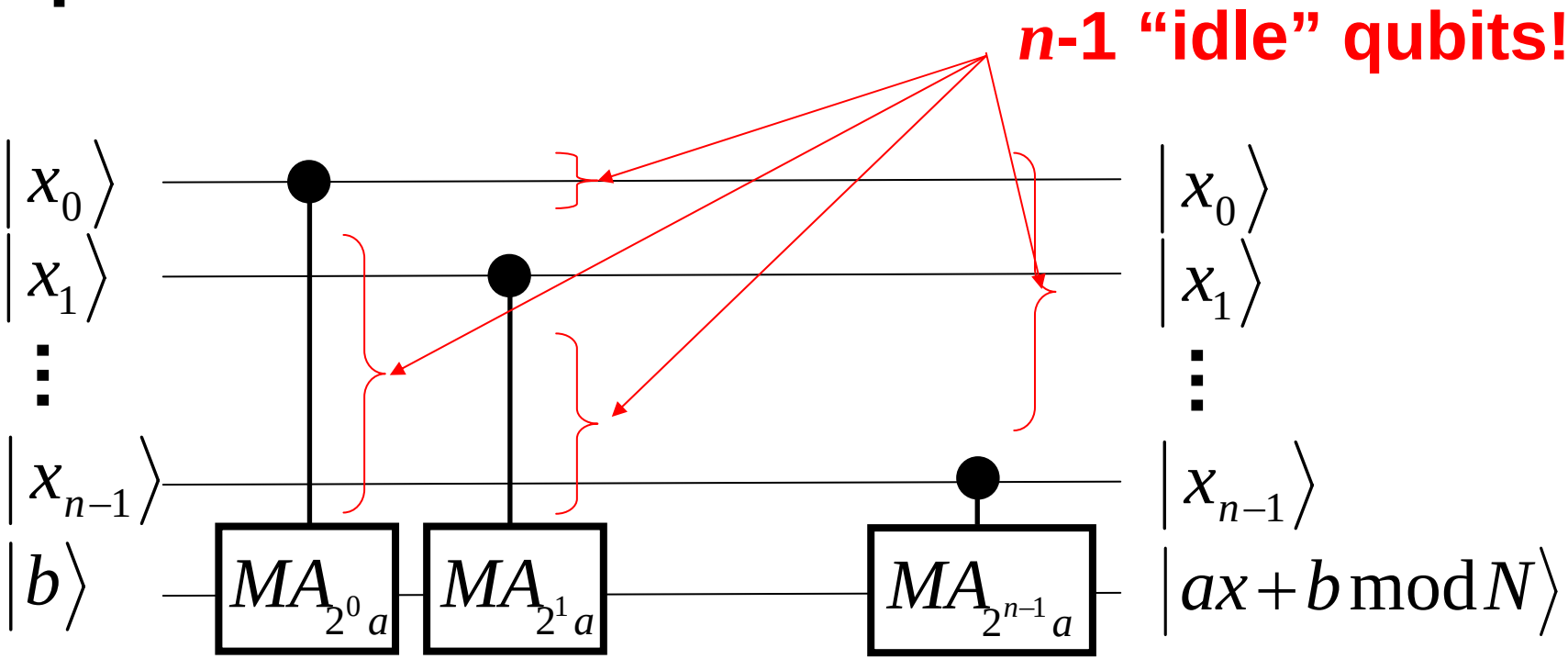
Modular addition

New circuit!



Many “idle” qubits

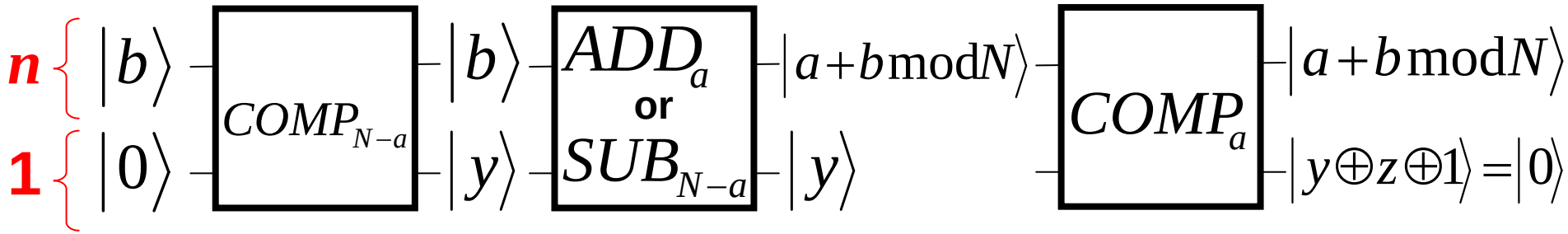
- **There are *n*-1 “idle” qubits while we perform a modular addition**



We use these “idle” qubits as “uninitialized” ancillary qubits

Our circuit for modular addition

- **For a classical number a , we construct a quantum circuit for modular addition that maps $|b\rangle$ to $|a+b \bmod N\rangle$**



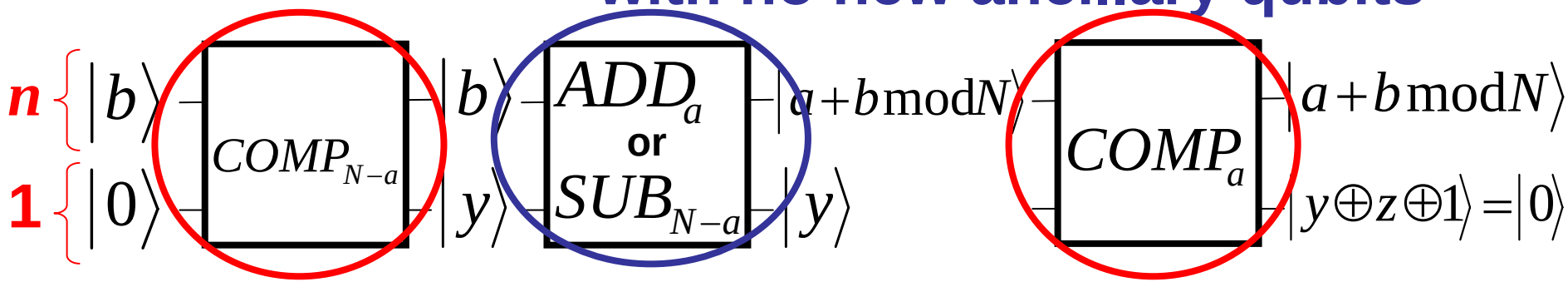
$$y = \begin{cases} 1 & a+b<N \\ 0 & otherwise \end{cases} \quad a+b \bmod N = \begin{cases} a+b & y=1 \\ a+b-N & otherwise \end{cases} \quad z = \begin{cases} 1 & a+b \bmod N < a \\ 0 & otherwise \end{cases}$$

$$a+b<N \Leftrightarrow a+b \bmod N \geq a \Rightarrow y=1 \Leftrightarrow z=0 \Rightarrow y \oplus z \oplus 1 = 0$$

Our and Beauregard's circuits

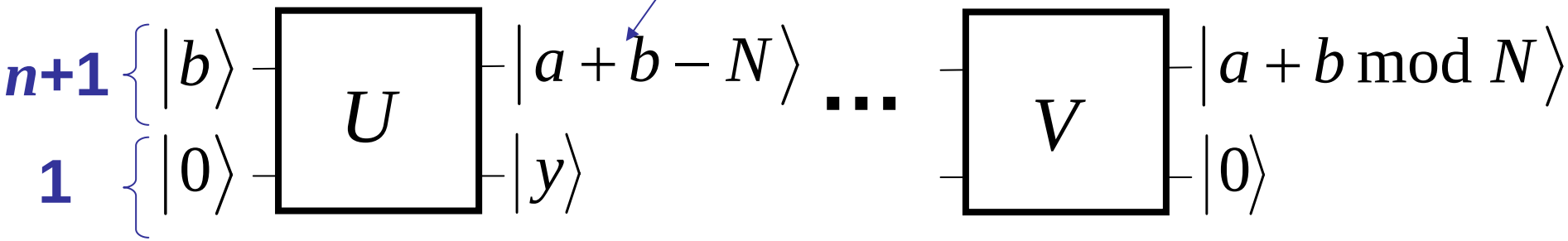
Our circuit

Draper's QFT-based addition with no new ancillary qubits



Beauregard's circuit

n+1 qubits



If we use no new ancillary qubits, the number of qubits is less than that in Beauregard's by one

Comparison

- **For a classical number a , we construct a quantum circuit for comparison that maps $|b\rangle|z\rangle$ to $|b\rangle|z \oplus y\rangle$, where**

$$y = \begin{cases} 1 & a > b \\ 0 & otherwise \end{cases}$$

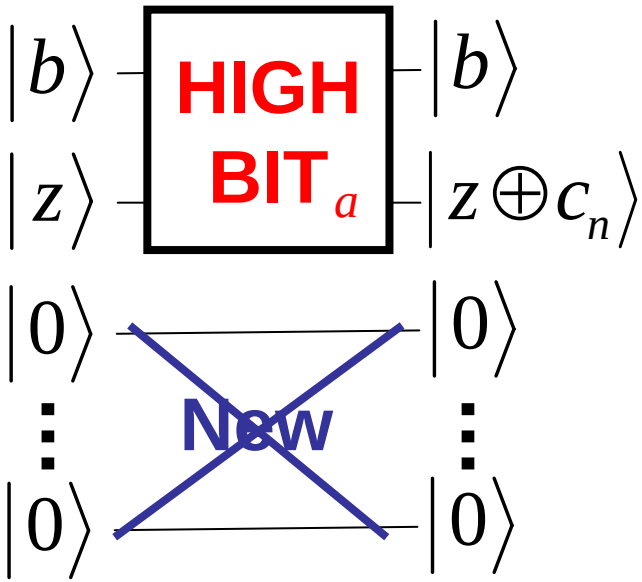
$$= \text{the high bit of } a + b'$$

We construct a circuit for computing only the high bit (the last carry bit) using the conventional adder

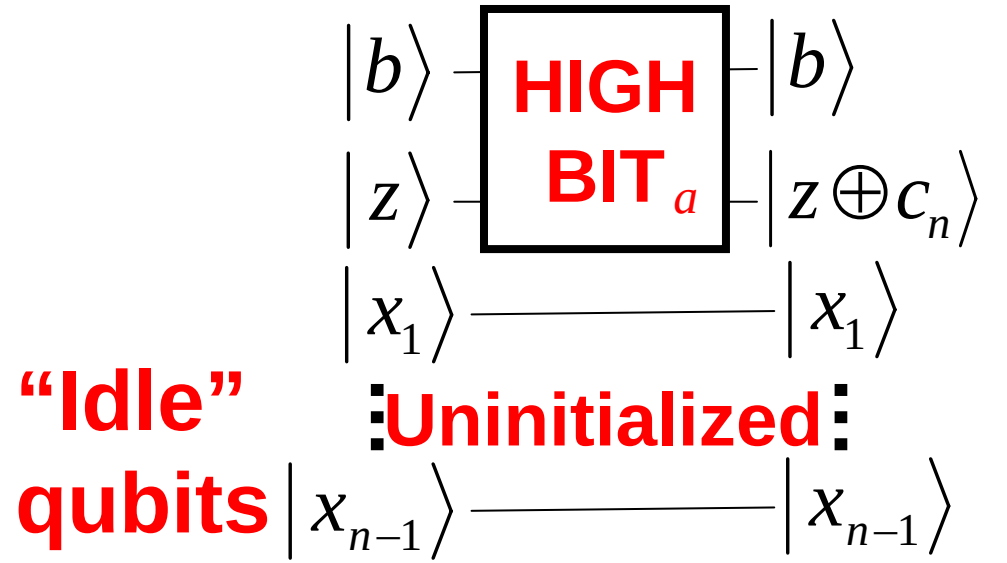
But, the adder uses n-1 new ancillary qubits

The use of “idle” qubits

- **We use not *n*-1 new ancillary qubits but “uninitialized” ancillary qubits**
- **They are available in the other register!**



The conventional adder



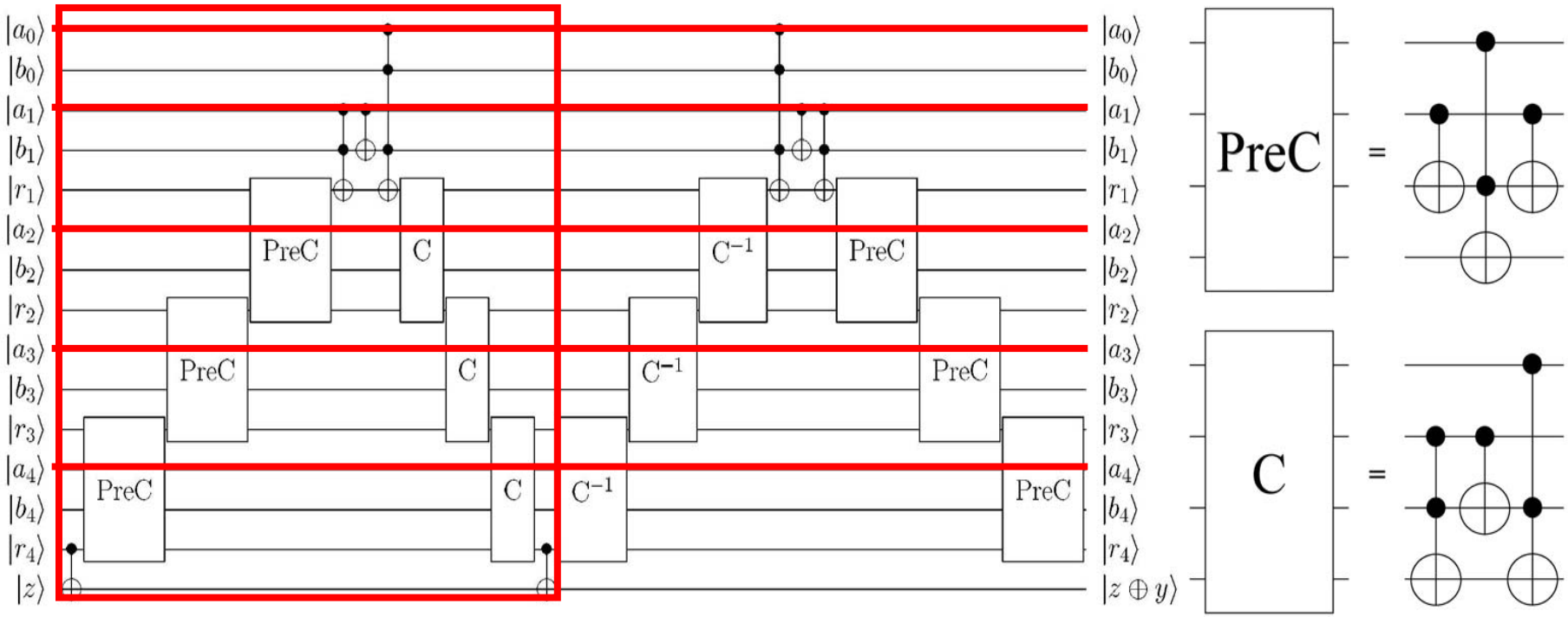
A modified adder

HIGHBIT gate

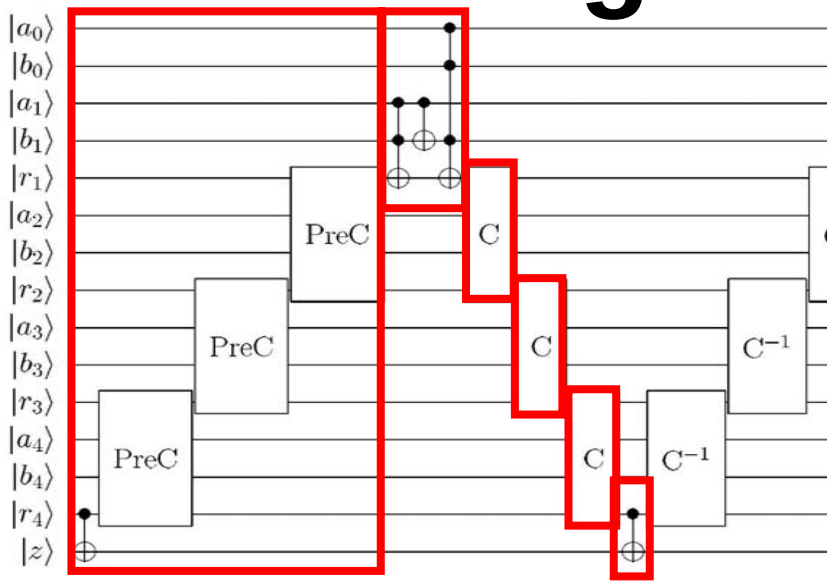
- **We construct a circuit for the operation**

$$\left|b\right\rangle\left|z\right\rangle\left|r_1\cdots r_{n-1}\right\rangle \mapsto \left|b\right\rangle\left|z\oplus c_n\right\rangle\left|r_1\cdots r_{n-1}\right\rangle$$

by modifying the conventional adder



Changes of the values



$$|b\rangle|z\rangle|r_1 r_2 r_3 r_4\rangle$$

$$\mapsto |b\rangle|z \oplus c_5\rangle|r_1 r_2 r_3 r_4\rangle$$

z	r_1	r_2	r_3	r_4
$\longrightarrow$ $z \oplus r_4$	r_1	$r_2 \oplus (a_2 \oplus b_2) r_1$	$r_3 \oplus (a_3 \oplus b_3) r_2$	$r_4 \oplus (a_4 \oplus b_4) r_3$
$\longrightarrow$ $z \oplus r_4$	$r_1 \oplus c_2$	$r_2 \oplus (a_2 \oplus b_2) r_1$	$r_3 \oplus (a_3 \oplus b_3) r_2$	$r_4 \oplus (a_4 \oplus b_4) r_3$
$\longrightarrow$ $z \oplus r_4$	$r_1 \oplus c_2$	$r_2 \oplus c_3$	$r_3 \oplus (a_3 \oplus b_3) r_2$	$r_4 \oplus (a_4 \oplus b_4) r_3$
$\longrightarrow$ $z \oplus r_4$	$r_1 \oplus c_2$	$r_2 \oplus c_3$	$r_3 \oplus c_4$	$r_4 \oplus (a_4 \oplus b_4) r_3$
$\longrightarrow$ $z \oplus r_4$	$r_1 \oplus c_2$	$r_2 \oplus c_3$	$r_3 \oplus c_4$	$r_4 \oplus c_5$
$\longrightarrow$ $z \oplus c_5$	$r_1 \oplus c_2$	$r_2 \oplus c_3$	$r_3 \oplus c_4$	$r_4 \oplus c_5$

Complexity analysis

- **Our circuit uses $2n+2$ qubits**
  - **The sequential computation of the QFT uses 1 qubit**
  - $|b\rangle, |z\rangle, |x\rangle$ **in modular multiplication use $2n+1$ qubits**
- **The size of our circuit is about half that of Beauregard's circuit**
  - **The number of QFT-based additions is about half that in Beauregard's circuit**

Conclusions

- **We construct a quantum circuit for order-finding using $2n+2$ qubits**
  - **The number of qubits is less than that in any other circuit ever constructed for order-finding**
- **The key ingredient of the circuit is the HIGHBIT gate that uses “uninitialized” ancillary qubits**